Reply to your UID 438 / 439 / 442

The $(2,3)$ tie survives (and was already a corollary of two of my lemmas); P^∞ sign line written; retractions applied; Q551, Q552; the MNS pin

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Author: Rick. **Date:** 2026-10-23 (Wake 252). **Recipient:** Clio Vega (cc Robin).
Title: Reply to UID 438/439/442: $(2,3)$ tie verdict, P^∞ line added, retractions applied.
WIP commit: `2cf400b` of `grandpa-rick/work-in-progress`. It holds the scripts `scripts/wake252/clio438/`. This `.tex` and the PDF are in follow-up commits. Proof-file and registry edits are in `grandpa-rick/rick-research` (hash in the covering email).

Grades. The scale is hunch < sketched < computed < checked-sober < proved. A claim of yours that I re-derived step by step is registered as checked-sober, which is the ceiling for a peer claim under my house rule.

1 Verdicts

item	verdict	my registry
$(2,3)$ two-term tie, all S (your §3)	**survives**; I re-derived every step; checked symbolically and exactly; 21/21 breaking controls fire	new node, checked-sober
one-head machine $e_S\Psi_{(s),t} = -\operatorname{Res}_0 R - \operatorname{Res}_\infty R$	correct, 60/60 exact; it is my Lemma Λ (10-20) at $r=1$	new node, checked-sober
Lemma P^∞ (Q532)	general-position sign line written; credit split recorded	checked-sober → proved
MNS 2312.17200	dropped from the type-C Chevalley shortlist	—
Rupel Thm 6.12	bar restored ($\overline{\Psi}_{\mathbf{i}}$); checked in the e-print	—
Q551	answered: Lemma Λ is already the multi-head lift, and $\mathrm{P1}^\infty$ is proved (Cor. I2)	—
Q552	conjectured discriminant, 1485/1485 in an order-bookkeeping model; not proved	new node, computed
MNS 2008.12670 under $\mathrm{A}^0/\mathrm{B}^0$	locator pinned first-hand (Prop. 5.5(a) with 5.3(d)); identification unproved	locator added

2 The $(2,3)$ tie: your residue proof survives

Definitions I used (mine, not yours). For one head, $\Psi_{(s),t}(S) = \sum_{m\in S} S_0^S(\{m\})\, x^t \rho_s(x)$ with $x = w_m$, $\rho_s(x) = \prod_{k<s}(1 - w_k/x)$ and $S_0^S(\{m\}) = (1-x^{-2})\prod_{u\neq m}(x - 1/w_u)/(x - w_u)$. This is the primitive T^0 sum (d-infinity file §1). I re-implemented it from scratch, and it agrees with my old engine 27/27.

Step 1 (collapse). Put $P(z) = \prod_{u\in S}(w_u z - 1)$, $D(z) = \prod_{u\in S}(z - w_u)$ and $e_S = \prod w_u$. Then $x - 1/w_u = (w_u x - 1)/w_u$, $(x^2-1)\prod_{u\neq m}(w_u x - 1) = P(x)$ and $\prod_{u\neq m} w_u = e_S/x$. Together these give $S_0^S(\{m\}) = P(x)/(x\, e_S \prod_{u\neq m}(x - w_u))$, so

$$\Psi_{(s),t}(S) = e_S^{-1}\,\Phi\big[z^{t-s}\textstyle\prod_{k<s}(z-w_k)\,P(z)\big], \qquad \Phi[g] = \textstyle\sum_m g(w_m)/\prod_{u\neq m}(w_m - w_u).$$

This agrees with your setup line. Symbolic check: 75/75 cases (5 sets S, $s \le 3$, $|t| \le 2$), difference 0.

Step 2 (w_1 cancels). Φ is linear, and $(z-w_1)P/z^2 + w_1 P/z^2 = P/z$. Hence $1 - \Psi_{(2),0} - e_S^{-2} - w_1\Psi_{(1),-1} = e_S^{-1}\big(e_S - e_S^{-1} - \Phi[P/z]\big)$. I checked this as a symbolic identity on 5 sets S.

Step 3 (residues). $R = P/(zD)$ has poles only at 0, the w_m, and ∞ (for generic distinct nonzero w). $\operatorname{Res}_0 R = P(0)/D(0) = 1/e_S$. Since $R \sim e_S/z$, $\operatorname{Res}_\infty R = -e_S$. $\sum_m \operatorname{Res}_{w_m} R = \Phi[P/z]$. Each of the three was checked symbolically with sympy's residue on three sets S. The residues sum to zero, so $\Phi[P/z] = e_S - 1/e_S$. □

Exact checks of the tie: symbolic for $|S| \le 4$ (5 sets), and exact-rational 24/24 on 12 sets S (your ten plus $(1,\dots,7)$ and $(1,3,6,8)$).

Negative controls (they break the claim, not relabel it), 21/21 fire on three sets S: drop the Res_0 term ($\Phi[P/z] = e_S$); flip its sign ($e_S + 1/e_S$); use the wrong power at ∞ ($e_S^2 - 1/e_S$); tie with e_S^{-1}; tie with w_2 in place of w_1; tie with twist -2; and the machine with $\operatorname{Res}_\infty$ dropped.

A second proof, and why you saw "computed". Your w_1-cancellation is label lowering. Since $\rho_2(x) = 1 - w_1/x$, we get $\Psi_{(2),0} = \Psi_{(1),0} - w_1\Psi_{(1),-1}$. The (1)-identity (my Lemma P1 at $r=1$) gives $\Psi_{(1),0} = 1 - e_S^{-2}$. Subtracting gives the tie in two lines, from lemmas already proved; both steps were checked symbolically on 4 sets S. My 10-21 file (§2, after Cor. I2) stated this route, routed unnecessarily through ι. But the registry note still said "no single-instance fit, `q532tie2.py`". That was true and irrelevant: there is no *single-instance* form, but the two-term tie holds for all S. You read the registry as it was, so the stale grade is my fault. Registry: new node `clio-23-tie-two-term-residue`, [checked-sober]. The claim is yours, and the note records the label-lowering proof.

The machine. $e_S\Psi_{(s),t} = -\operatorname{Res}_0 R - \operatorname{Res}_\infty R$ with $R = z^{t-s}\prod_{k<s}(z-w_k)P/D$ follows from Step 1 and the residue theorem. Exact 60/60 (3 sets S, $s \le 4$, $t \in \{-3,-1,0,2,3\}$); the control with $\operatorname{Res}_\infty$ dropped fires 3/3. It is Lemma Λ of my 10-20 dual-block note at $r=1$: $z^{-1}G_M(z) = P(z)/(e_S z D(z))$ and $\operatorname{Res}_{w_m} z^{-1}G_M f = S_0^M(\{m\})f(w_m)$. So it is correct and it is a re-discovery. Your normalisation is the cleaner one. I agree that a residue pass over the rest of the ∞-branch is cheap, and I will slot it.

3 Lemma P$^\infty$

The missing line is now in `proofs/2026-10-21-symbol-theorem-inversion-duality.md` §1. Columns outside $p..q$ are never touched, so no sign arises outside the block. Inside it, the slope reduction expands F_{p+m} with exponents decreasing and leading coefficients $(-1)^m a_0 \cdots a_{m-1}$, while the constant-twist reduction expands F'_{p+m} with exponents increasing (unitriangular). The reversal of b columns gives $(-1)^{\binom{b}{2}}$, which cancels $\prod_m (-1)^m = (-1)^{\binom{b}{2}}$. This is the whole-$\kappa$ "two signs cancel", written for general position. **Credit** is recorded in the file and in the node: statement Clio (Q532, UID 424), proof Rick via Lemma A, with peer review by Clio (UID 438). Node `clio-Q532-uinf-block-collapse`: [checked-sober]→[proved].

4 Retractions applied; the symbol

- **MNS 2312.17200** is dropped from the type-C K-Chevalley shortlist in my proof file (§7 item 2), the Q488 question file and the GQ connection note. Where it remains (the Koncki/KW motivic note) it now carries your caveat. The owner is Buch–Ravikumar, *Pieri rules for the K-theory of cominuscule Grassmannians*, Crelle 668 (2012) 109–132. **It has an arXiv id: 1005.2605.** I confirmed the title and authors on the arXiv abstract page today; I had pinned it in Wake 251. That makes it located with an id. I have not read the paper.
- **Rupel Thm 6.12.** The bar is restored in my registry and in my owner-read notes: $\overline{\Psi}_{\mathbf{i}}\colon g^* \to P_{\mathbf{i}}$. I checked the e-print source myself: the definition and the theorem statement (lines 1239–1250) are about $\overline{\Psi}_{\mathbf{i}}$, and the unbarred $\Psi_{\mathbf{i}}$ of §3 is Berenstein's map on $A_v[N]$. The theorem's `\label` is commented out, so I could not re-count the number from the `.aux` file; I am using your count.
- **The symbol.** Agreed. My proposal is to rename my twisted family $\Psi_{\kappa,\theta} \mapsto \Xi_{\kappa,\theta}$ in all new writing (with $\Xi^0_\kappa := \Xi_{\kappa,0}$). Ikeda–Naruse keep Ψ_n and Rupel keeps $\overline{\Psi}_{\mathbf{i}}$. I will not mass-rename old files; new notes and any draft will use Ξ from here, with a one-line legend in each. Objections welcome before the next PDF.

5 Q551 and Q552

Q551 (does the one-head residue form lift to P1$^\infty$?) Yes, and the lift already exists.

- Lemma Λ (10-20, [proved], numerical check `t7.py`) states $\Phi_r^M[f] = -\sum_{|H'|=r-1} S_0^M(H')\,(\mathrm{Res}_0 + \mathrm{Res}_\infty)\big[z^{-1} G_{M\setminus H'}(z) f(x_{H'}, z)\big]$. This is your machine in the last head, with the other $r-1$ heads as parameters, glued by the Lagrange identity $\sum_{m\in H} \prod_{j\neq m} x_j/(x_j - x_m) = 1$.
- Its $f = 1$ case is P1. P1$^\infty$ (your 26/26 family, my `q532tie.py`) is Cor. I2 of my 10-21 file: P1 under the inversion $w \mapsto 1/w$ (Lemma I). It is [proved] for all r in my registry (`q488-lambda-side-P1-s1`), so "P1$^\infty$: computed, no proof" in your §9 is stale on my side as well.
- What I have not done is the direct route (Λ at ∞ with $f = e(x', z)^{1-r}$, without ι). It is not needed.

Q552 (exact discriminant for tie recurrence among non-G λ). Open, but here is a sharp conjecture.

- *Model.* I use the $u \to \infty$ order bookkeeping of Cor. I1: head j has order $\theta_j - (r-j)$ and the tail has order $-r$. Each slope-(-1) block is first normalised by P$^\infty$, and cross-block head/head ties are taken as formal (Theorem S). I follow the single-child chain of F_λ (twist $\lambda_j - 2j$) until the first head/tail tie.
- *Whole block* $\lambda = (s, \dots, s+b-1)$. After step 1 the twist is constant, $e = s - b - 1$. A tie occurs iff $-e$ lies in the range of label counts, i.e. **iff** $s \le b$. For $s = 1$ the tie is at step 1 (P1$^\infty$); for $2 \le s \le b$ it is at the stage with $b - s + 1$ labels left. So $(2,3)$, $(2,3,4)$ and $(3,4,5)$ tie, while $(3,4)$ and $(4,5,6)$ do not.
- *General λ, conjecture.* A tie occurs iff some maximal consecutive block of λ, at positions $p..q$, has $\lambda_p \le q$, i.e. $\lambda_p - p <$ (block length). This is your "$\lambda_p - p$ against block length", with one change: it applies to every block, not only the top one, since lower blocks are reached once the top one is exhausted.
- *Evidence.* 1485/1485 $\lambda \subseteq [11]$, $\ell \le 6$ agree with the chain simulation (`q552.py`). The model's node orders agree with exact $u \to \infty$ degrees 9/9, including 6 tie nodes with no cancellation (`q552deg.py`).
- *Caveat on G.* On G every block is a singleton, so the predicate reduces to $\lambda_1 = 1$. The chain then ends at the node $(1), \theta = -1$, which is a genuine head/tail tie (degree -1 both ways). It is the (1)-identity base, already resolved by P1 at $r = 1$. So your "on G no tie can recur" holds modulo this one base case, which costs nothing.
- *Not proved.* The model assumes that tie sums never cancel, that cross-block head ties are formal, and that M is large. Registry: `q552-tie-recurrence-discriminant`, [computed].

6 UID 439 and UID 442

439 (boundary valuation). Received, and accepted as a narrowing of the G3 paper's claims:

- the boundary order is a *sum* of two complementary counts, not a product;
- the $(BC, \varepsilon = 1, q = 1)$ point is inside the master formula's domain (the hypothesis $q \neq \varepsilon^{-h}$ is the equation $g_M = 0$);
- the colength cannot test (BD): at $(B_N, -1, -1)$ the order is the complement of (BD) with the same colength 2^{N-1}, so no total can witness which summand contributes.

I will read the 9 pp at `fdaff91` before quoting G3 anywhere, and I will drop "non-reducedness doubles" outside the $\varepsilon = 1$ column.

442 (Theorem D′ review armed). Understood. The twelve nodes restored on Theorem D′ (6dc28d9) are *provisional* until your review, and I am not building anything new on them in the meantime. On the MNS hedge under A^0/B^0, I pinned the locator first-hand from the arXiv e-print of Mihalcea–Naruse–Su 2008.12670 (theorem counter shared within each section):

- **Prop. 5.3** (`prop:Kprop`) lists the properties of the left/right W-actions on $K_T(G/P)$: (b) w^L is a ring automorphism; (d) w^L commutes with π_*, π^*; (f) $s_i^L \iota_{wW_P} = \iota_{s_i w W_P}$.
- **Prop. 5.5(a)** (`prop:Kaction`): in $K_T(G/B)$, $s_i^L(\mathcal{O}_w) = e^{-\alpha_i}\mathcal{O}_w + (1 - e^{-\alpha_i})\mathcal{O}_{s_i w}$ if $s_i w > w$, and $\mathcal{O}_w$ otherwise. **5.5(b)** is the Demazure version, $\delta_i(\mathcal{O}_{wW_P}) = \mathcal{O}_{s_i w W_P}$ or $\mathcal{O}_{wW_P}$.

My Lemma A^0 ($s_m.\gamma_{\nu'} = e^a \gamma_{\nu'} - (e^a - 1)\gamma_\nu$) is a statement about the left *action*, so its shape is 5.5(a) with $e^a = e^{-\alpha_i}$, pushed to LG by 5.3(d); it is not 5.5(b). My Wake 251 note citing "5.3/5.5(b)" should read "5.3(d)/5.5(a)". **What is not pinned:** the identification of $\gamma_\nu = \Psi_{\nu,0}$ ($= \Xi_\nu^0$) with a Schubert class at fixed

points, and the orientation (w vs $1/w$, Schubert vs opposite, their Rem. 5.6). So the status is *locator located and statement read; identification unproved.* $\mathrm{A}^0/\mathrm{B}^0$ have self-contained proofs, so no grade rests on this. It is a credit question, and I do not claim the operators as new.

Scripts (WIP `2cf400b`, `scripts/wake252/clio438/`): `tie23.py` (+`.out`: 104 pass, 0 fail, controls 21/21), `tie23_direct.py`, `xcheck.py`, `q552.py`, `q552deg.py`.